INTERIM MILITARY SPECIFICATION

CENTRIFUGE, LABORATORY, REFRIGERATED, 120/230 volt, 50/60 Hz, AC

This limited coordinated military specification has been prepared by the Defense Personnel Support Center, based on currently available technical information. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers requirements for a floor model refrigerated centrifuge suitable for use in separating whole blood into plasma, serum and packed cells, and platelet rich plasma into plasma and the platelet concentrates.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

(DD Form 1426 to be included in the final document.)

AMSC/NA FSC 6640

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SPECIFICATIONS

FEDERAL

Military

MIL-D-42048 - Deployable Medical Systems, General Requirements for.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

3. REQUIREMENTS

3.1 General. The complete requirements for acquiring the refrigerated centrifuge described herein shall consist of this specification and the latest issue of Military Specification MIL-D-42048, except where otherwise indicated.

3.2 Design and construction.

3.2.1 General. The centrifuge shall the manufacturer's latest design and shall be fabricated using materials and construction techniques that are consistent with similar items and techniques that are commercially available. The design and the placement of the component parts of the centrifuge shall be that best suited to provide a rugged, durable, efficient and quiet operating centrifuge.

3.2.2 Design. Shall be a floor mounted refrigerated centrifuge for separating whole blood into plasma, serum, and packed cells. As a minimum, shall consist of the following:

(a) A floor mounted cabinet
(b) A rotor chamber
(c) A centrifuge motor
(d) A drive train
(e) A rotor assembly

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(f) A refrigeration system
(g) Controls
(h) Meters
(i) Indicator lights
(j) Power Cord
(k) Accessories

3.2.3 Circuitry. Solid-state electronic circuitry shall be incorporated into the design and manufacturing of the centrifuge wherever practicable.

3.2.4 Component Units

3.2.4.1 Cabinet. The cabinet shall provide the housing for the centrifuge, and shall be designed to have the strength and stability to support all the centrifuge and refrigeration functional (component) units, and to withstand all stresses encountered in normal usage. Access for servicing the components (within the cabinet) shall be provided. The cabinet shall be supported on four ball-bearing, swiveling casters having sufficient load-carrying capacity. The housing shall include a hold down bracket for securing the centrifuge (cabinet) to the floor.

3.2.4.1.1 Cabinet cover. The top of the cabinet shall allow for access to the centrifuge chamber, and shall be provided with a hinged and gasket-seated cover. The cover shall have a positive and easy operating lock. The electric/electronic circuitry design shall include an interlock system that prevents the rotor from spinning when the cover is open, and prevents the cover from being opened when the rotor is still spinning. The cover shall have sufficient weight and strength to withstand, without penetration or breakage, any impact occasioned by failure of the rotating components within the centrifuge chamber. The cover shall incorporate a window through which independent rotor speed measurements may be made by using a phototachometer.

3.2.4.1.2 Ventilation. The cabinet housing shall be provided with openings that insure ample ventilation for the centrifuge motor and refrigeration unit condenser. A hot-air exhaust shall be located at the rear of the cabinet to permit side by side operation of the centrifuge with other refrigerated centrifuges.

3.2.4.1.3 Instrument panel. The cabinet shall have one or more instrument panels to accomodate the dials, control knobs, switches, timers, indicator lights, meters, etc.

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3.2.4.2 Rotor chamber. The rotor chamber shall be constructed with a stainless steel inner lining and a steel safety containment jacket capable of withstanding without penetration any impact occasioned by the failure of rotating components. The cooling coils shall be sandwiched between these layers, and insulated in such a way that only the stainless steel lining must be cooled to bring the chamber to the selected temperature. The inside volume of the chamber shall provide free swinging of all approved accessories. The centrifuge chamber and its cover shall be sufficiently insulated to minimize the transfer of heat from the room to the centrifuge chamber, and also, to prevent vapors from condensing on the centrifuge cabinet.

3.2.4.3 Centrifuge motor. The centrifuge motor shall be a 1.86 Kw and 2.5 hp rated gyro-action (high torque) direct drive type DC motor. In addition, the motor shall be a continuous duty, variable-speed motor, having high torque characteristics for smooth, quick rotor acceleration. Motor bearings shall be sealed, permanently lubricated, and replaceable by qualified personnel. The motor shall be installed in a manner which minimizes vibration, affords protection from spillage of substance being tested, and facilitates replacement of contact brushes.

3.2.4.4 Drive train. The drive train shall be so designed that there exists a direct coupling between the rotor and the motor. Belts or chains shall not be allowed. Shafts, bearings and couplings shall be capable of supporting the static and dynamic forces encountered in their operation, shall be permanently lubricated, and shall be replaceable by qualified personnel. The drive assembly shall be capable of absorbing a rotor imbalance of up to 10 grams without damage to the rotor shaft.

3.2.4.5 Rotor assembly. The centrifuge shall be provided with a six place centrifuge rotor assembly. The rotor assembly shall consist of a rotor body, buckets, wind screen and wind screen cover, all fabricated of appropriate materials. The rotor shall be able to be easily inserted and locked onto the centrifuge shaft. Buckets shall be of free swinging type and have a capacity of one liter. The wind screen cover shall be attached to the center axle or support of the rotor, and shall cover the circumference of the rotor and bucket assembly to minimize wind drag. The centrifuge rotor shall be capable of sustained, fully loaded operation at a speed of 4200 RPM with a Relative Centrifugal Force (RCF) of 5010g.

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3.2.4.5.1 Rotor stabilizer. The centrifuge shall be provided with a rotor stabilizer that allows the centrifuge to be moved without damage to the rotor.

3.2.4.6 Refrigeration system. The refrigeration system shall be of the hermetically-sealed type, consisting of a motor-driven compressor, condensing unit, expansion valve, evaporator, tubing and other accessory components. The refrigeration unit shall be supported within the centrifuge cabinet in a manner so as to prevent the transmission of vibration. The unit shall be compact in design and efficient in operation and controlled with solid state temperature sensing circuitry. The refrigeration system shall provide the capability to maintain the rotor chamber at 4°C during full speed operation at an ambient temperature of 25°C for runs up to one hour in duration.

3.2.4.7 Controls. The refrigerated centrifuge shall incorporate the following controls which shall be conveniently placed and clearly marked on the cabinet housing:

(a) Power ON/OFF switch: to activate and deactivate all circuits.
(b) Start switch to initiate a run.
(c) Stop switch to terminate a run.
(d) Time Control: continuously adjustable in increments of one minute from 0 to not less than 30 minutes, and having a "hold" selection for runs of indefinite duration.
(e) Speed Control: continuously adjustable with settings ranging from zero to 6000 revolutions per minute (RPM).
(f) Automatic Rate Control: provides variable, operator selected, automatic control of the rate of acceleration up to 250 RPM, and rate of deceleration below 500 RPM. "Locks out" the compressor below 250 RPM during acceleration and deceleration to prevent "shudder" to the rotating samples.
(g) Brake Control: provides variable, operator selected, automatic control of the braking rate.
(h) Temperature Control: provides operator selection of the temperature of the rotor chamber with an accuracy of plus or minus 1°C. An automatic over-temperature control shall automatically terminate a run if the chamber temperature exceeds the set temperature by 7°C and greater. An over-temperature indicator light shall indicate that the run has been terminated.
(i) Integrator: to accurately calculate the total centrifugal effect during centrifugation which provides true run to run reproducibility.

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3.2.4.8 Meters The refrigerated centrifuge shall incorporate the following meters:

(a) Tachometer: to indicate rotor rotation. The tachometer shall have a range of 0 to 6000 RPM with an accuracy of ± 2% at 6000 RPM and shall be maximally graduated in increments of 100 RPM if analog. A digital readout tachometer comparable range and accuracy shall be acceptable.

(b) Thermometer: to indicate temperature within the rotor chamber. The thermometer shall have a scale ranging from -20 to +40 degrees Celsius (C), and shall be maximally graduated in 2 degree increments. A digital readout thermometer of comparable range shall be acceptable.

3.2.4.9 Indicator lights. The refrigerated centrifuge shall incorporate the following indicator lights:

(a) Power: to indicate that electrical circuits are energized.
(b) Brake: to indicate that braking is in progress.
(c) Door: to indicate that rotor has stopped and door may be safely opened.
(d) Imbalance: to indicate that rotor load is not evenly distributed. A potentially harmful imbalance will terminate the cycle.
(e) Over temperature: to indicate that temperature in chamber exceeds set temperature by at least 7 degrees (C).
(f) Brushes: to indicate that motor contact brushes are worn and in need of replacement.

3.2.4.10 Power cord. The power cord and plug shall conform to requirements contained in MIL-D-42048.

3.2.4.11 Accessories. The following accessories shall be supplied with the refrigerated centrifuge:

(a) A pair of replacement electrical carbon contact brushes for the centrifuge motor.

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(b) A package of six (6) rubber balancing discs for use with the 1 liter buckets of the rotor assembly. A disc shall be suitable for adding to one of the buckets in order to equally balance a pair of opposite buckets on the rotor assembly.

3.3 Safety features. The centrifuge shall be so constructed as to furnish adequate protection to the user both from electrical shock and from spillage or breakage or operational parts or objects being rotated at high speeds by the centrifuge. An interlock system shall be included to prevent the rotor from spinning when the door is open and to prevent the door from being opened when the rotor is still spinning.

3.4 Performance. The refrigerated centrifuge shall be capable of operation from 110/220 volt, 50/60 Hz AC supply lines, in accordance with dual electric requirements in MIL-D-42048.

3.4.1 Operation. When tested as specified in 4.4.1, the refrigerated centrifuge shall demonstrate the proper functioning of mechanical, electrical and refrigeration components without any evidence of or tendency to failure.

3.4.2 Rotor speed. When tested as specified in 4.4.2, the refrigerated centrifuge shall be capable of operating within the specified speed ranges. The tachometer speed readings shall be within the accuracy requirements of paragraph 3.2.4.8.

3.4.3 Refrigeration. When tested as specified in 4.4.3, the blood samples after centrifuging shall not exhibit any evidence of the following:

(a) Formation of ice crystals.
(b) Sample(s) reaching a temperature greater than 10 degrees Centigrade.
(c) Sample(s) being hemolyzed.

3.5 Reusable container. The reusable container requirements in MIL-D-42048 shall not apply to this specification.

3.6 Workmanship. The refrigerated centrifuge shall be free from defects which detract from its appearance or impair its serviceability.

3.7 Unit. Each (EA). One refrigerated centrifuge, as specified, constitutes one unit.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performances of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 First article inspection. First article inspection shall be performed on the refrigerated centrifuge when a first article sample is required. This inspection shall include examination and tests.

4.1.4 Quality conformance inspection. Quality conformance inspection shall include the examination and tests specified herein.

4.1.5 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, and the lot size.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I. The unit of product for sampling purposes shall be one refrigerated centrifuge.

TABLE I. Sampling for examination.

	Inspection level	AQL (percent defects)
For visual examination		
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted as follows:

a. First article. Where first article sample is required, all tests shall be performed on that unit.

b. Acceptance testing. When a first article sample has not been required, one unit shall be subjected to all tests specified herein. If a first article sample has successfully passed all tests, performance testing shall not be required at acceptance.

4.3 Examination. The refrigerated centrifuge shall be examined to determine compliance with the requirements of this specification in accordance with the classification of defects in table II.

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TABLE II. Categories and Defects*

Categories	Defects
Major	
101	Unit not complete with all component parts.
102	Unit not free of wobbling or rocking when placed on a flat horizontal surface.
103	Unit not free of sharp edges or burrs.
104	Components not usable or properly secured.
105	Power cord not free of cuts, tears, or abrasion.
106	Power cord not properly secured to cabinet.
107	Blades of power cord not properly aligned.
Minor	
201	Unit not supplied in a clean condition.
202	Unit not free of chips, scratches, or other appearance defects.
203	Electric control markings not complete, legible, or permanent.
204	Unit not supplied with two copies of operator and service manuals.

* Examination shall not be limited to the classified possible defects listed herein.

4.4 Tests. Tests shall be conducted to determine compliance with this specification.

4.4.1 Operational test. All refrigerated centrifuges shall be tested for compliance with performance requirements as follows:

Each refrigerated centrifuge shall be connected to a suitable power source, in accordance with the manufacturer's instructions, and operated to demonstrate the proper functioning of mechanical, electrical, and refrigeration components. Any evidence of or tendency to failure of any component shall be cause for rejection (see 3.4.1).

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4.4.2 Rotor speed. The rotor assembly shall be fully assembled, in accordance with 3.2.4.5, and secured to the centrifuge shaft. The centrifuge shall be connected to a suitable power source and operated within each of the following three ranges:

150 to 500 RPM
2500 to 4000 RPM
5000 to 6000 RPM

Within each speed range, a measurement shall be taken with a calibrated phototachometer and compared with the speed setting on the speed control and the speed indication on the tachometer. A centrifuge that fails to attain a set speed, or that shows a variance of more than 120 RPM from the tachometer reading shall be cause for rejection (see 3.4.2). Sampling shall be conducted in accordance with MIL-STD-105, with an S-1 inspection level, 0.65 AQL and an acceptance number of zero.

4.4.3 Refrigeration test. A centrifuge shall be tested to determine the ability of its refrigeration system to lower and maintain a proper temperature for blood samples in the rotor chamber during centrifuging. A fully assembled rotor assemble (see 3.2.4.5) shall be secured to the centrifuge shaft. With the centrifuge buckets empty, the centrifuge shall be connected to a suitable power source and the temperature control set for +5 degrees centrifuge. After the centrifuge rotor chamber has reached +5 degrees centrifuge, blood samples at room temperature (e.g. 20 degrees Centigrade) shall be placed in the centrifuge buckets. The centrifuge shall be operated at 5000 RPM for a 20 minute period. After centrifuging, the samples shall be removed and checked to determine the following:

a. Whether samples have been cooled to such a state that ice crystals may have been formed in any of the samples.

b. Whether the temperature of any of the blood samples has not been lowered to at least 10 degrees Centigrade.

c. Whether any of the blood samples has been hemolyzed.

Any evidence of blood samples freezing or exceeding 10 degrees centigrade to exhibit hemolysis shall be cause for rejection of the centrifuge (see 3.4.3). One centrifuge from each lot of centrifuges shall be tested. The acceptance number shall be zero.

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5. PACKAGING

5.1 General. The complete requirements for packaging shall consist of this specification and the latest issue of Military Specification MIL-D-42048 except where otherwise indicated.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) National Stock Number (NSN).
(c) Selection of applicable levels of packaging and packing (see 5.1)
(d) Exception(s) to requirements in paragraph 3.1 of this specification.

6.2 Stock listing. This specification covers the following item listed in the Federal Supply Catalog:

NSN	ITEM IDENTIFICATION
6640-01-139-7457	CENTRIFUGE, LABORATORY, Refrigerated, 120/230 volt, 50/60 Hz, AC

6.3 Qualification. This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet military requirements.

CUSTODIANS:
Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

AGENT:

DLA-DM

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CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
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FDA-MPQAS

Project No. 6640-1493

Location: Disk # 11

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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